

5(a)(i)	<i>assimilates are sucrose, amino acids, (named) monosaccharides or other named disaccharide, (named) hormones, accept (named correct) mineral ions</i> <i>root as source</i> <i>idea that root provides (named) assimilate(s) for other parts of the plant ;</i> R if starch is included <i>root as sink</i> root is for storage / root is a storage organ ; A root receives (named) assimilates to, store / grow <i>ref. to bidirectional transport in phloem ;</i> direction of movement changes with demands of plant for (named) assimilates ;	2
5(a)(ii)	sucrose <u>and</u> amino acids ; A other named monosaccharides or disaccharide, (named) hormones, (one named correct) mineral ion(s) / organic acid(s) / sugar alcohol(s) / protein(s) / RNA / AW	1

Question	Answer	Marks
5(b)	any four from: 1 (assimilates) diffuse / enter / AW, into (phloem) sieve, tube / element, from companion cells / via <u>plasmodesmata</u> ; 2 water potential in sieve tubes (in source), decreases / is lower ; R 'lowers water potential gradient' 3 water enters (from surrounding tissues), by osmosis / down water potential gradient ; 4 hydrostatic pressure builds up ; A hydrostatic pressure difference created / <i>ref. to</i> hydrostatic pressure gradient 5 <u>mass flow</u> ; 6 unloading at, (named) sink, decreases / lowers, hydrostatic pressure ; 7 description of unloading as assimilates moving out of sieve tubes and water following ; I activity within companion cell	4

Question	Answer	Marks
5(c)	any three from: 1 TMV maximum accumulation is delayed in V1 and V2 / AW ; 2 <i>data to support mp1:</i> V1 does not start to increase, until day 9 / steeply until day 16 or V2 does not start to increase, until day 22 / steeply until day 24 ; 3 rate at which accumulation increases is slower than the control ; ora I ref. to time A ref. to steepness of accumulation in V1 and V2 compared with C 4 V1 and V2 (and C) reach maximum accumulation / AW ; 5 low quantities of PME do not decrease maximum TMV accumulation or normal PME quantity not required to reach maximum TMV accumulation ; 6 AVP ; for any other valid comparison, e.g. all are slow to start and then very quick to reach 100% all have the same, pattern / trend A correct data: C took 8 days, V1 took 25 days, V2 took 36 days (from infection) to reach maximum	3